

# Quotient Laws of Exponents

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## MA.7.NSO.1.1

I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

# Warm-Up

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1. Write as one power:  $3^2 \cdot 3^3$
2.  $\sqrt[3]{-27}$
3. Solve  $2x + 3 < 11$
4. Simplify  $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

# Warm-Up

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1. Write as one power:  $3^2 \cdot 3^3$   $3^5$

2.  $\sqrt[3]{-27}$   $-3$

3. Solve  $2x + 3 < 11$   $x < 4$

4. Simplify  $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$   $5$

# Notes

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## I. Quotient of powers — same base, subtract the exponents

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$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

Quotient of powers:  $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values:  $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

# Notes

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## II. Power of a quotient — the exponent goes top and bottom

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$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

Power of a quotient:  $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way:  $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$  and  $(0.8)^3 = 0.512$

# Notes

## III. The zero exponent — two roads to the same answer

| $2^4$ | $2^3$ | $2^2$ | $2^1$ | $2^0$ |
|-------|-------|-------|-------|-------|
| 16    | 8     | 4     | 2     | 1     |

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

Zero exponent:  $a^0 = 1$  for every nonzero  $a$

# Example 1

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$$\frac{(-2)^6}{(-2)^3}$$

**Which law? Then find the value.**

# Example 1

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$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3$$

$$(-2)^3 = -8$$

$$64 \div (-8) = -8$$

**Answer: -8**

# Your Turn

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$$\frac{3^6}{3^2}$$

**Which law? Then find the value.**

# Your Turn

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$$\frac{3^6}{3^2}$$

Which law? Then find the value.

$$\frac{3^6}{3^2} = 3^{6-2} = 3^4$$

$$3^4 = 81$$

**Answer:  $3^4 = 81$**

## Example 2

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$$\left(\frac{0.2}{10}\right)^4$$

**Find the value.**

## Example 2

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$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000}$$

$$\frac{0.0016}{10000} = 0.00000016$$

$$(0.02)^4 = 0.00000016$$

**Answer: 0.00000016**

# Your Turn

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$$\left(\frac{3}{4}\right)^3$$

**Find the value.**

# Your Turn

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$$\left(\frac{3}{4}\right)^3$$

Find the value.

$$\left(\frac{3}{4}\right)^3 = \frac{3^3}{4^3} = \frac{27}{64}$$

**Answer:**  $\frac{27}{64}$

# Whiteboards • Question 1 of 9

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$$\frac{10^5}{10^3}$$

**Answer it.**

# Whiteboards • Question 1 of 9

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$$\frac{10^5}{10^3}$$

$$\frac{10^5}{10^3} = 10^{5-3} = 10^2$$

**Answer:  $10^2$**

# Whiteboards • Question 2 of 9

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A website puts  $8^4$  photos into 8 equal albums.

The site is 30 days old.

**How many photos are in each album?**

**Answer it.**

# Whiteboards • Question 2 of 9

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A website puts  $8^4$  photos into 8 equal albums.

The site is 30 days old.

**How many photos are in each album?**

$$8^4 \div 8^1 = 8^{4-1} = 8^3$$

$$8^3 = 512$$

**Answer: 512**

# Whiteboards • Question 3 of 9

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$$\frac{(-6)^6}{(-6)^5}$$

**Answer it.**

# Whiteboards • Question 3 of 9

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$$\frac{(-6)^6}{(-6)^5}$$

$$\frac{(-6)^6}{(-6)^5} = (-6)^{6-5} = (-6)^1 = -6$$

**Answer: -6**

# Whiteboards • Question 4 of 9

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$$\left(\frac{-5}{10}\right)^7$$

**Answer it.**

# Whiteboards • Question 4 of 9

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$$\left(\frac{-5}{10}\right)^7$$

$$\left(\frac{-5}{10}\right)^7 = \frac{(-5)^7}{10^7}$$

**Answer:**  $\frac{(-5)^7}{10^7}$

# Whiteboards • Question 5 of 9

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$$9^0$$

Which is the value?

**A.** 1

**B.** 9

**C.**  $1/9$

**D.** 0

**Answer it.**

# Whiteboards • Question 5 of 9

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$$9^0$$

Which is the value?

Any nonzero number to the power 0 is 1.

$$9^0 = 1$$

**Answer: A — 1**

# Whiteboards • Question 6 of 9

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$$\frac{4^8}{4^2}$$

Which expression is equivalent?

**A.**  $1^6$

**B.**  $4^{10}$

**C.**  $4^6$

**D.**  $4^4$

**Answer it.**

# Whiteboards • Question 6 of 9

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$$\frac{4^8}{4^2}$$

Which expression is equivalent?

$$\frac{4^8}{4^2} = 4^{8-2} = 4^6$$

**Answer: C —  $4^6$**

# Whiteboards · Question 7 of 9

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$$\frac{0.5^8}{0.5^5}$$

**Answer it.**

# Whiteboards • Question 7 of 9

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$$\frac{0.5^8}{0.5^5}$$

$$\frac{0.5^8}{0.5^5} = 0.5^{8-5} = 0.5^3$$

$$0.5^3 = 0.125$$

**Answer: 0.125**

# Whiteboards • Question 8 of 9

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$$\left(\frac{10}{11}\right)^0$$

**Answer it.**

# Whiteboards • Question 8 of 9

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$$\left(\frac{10}{11}\right)^0$$

Any nonzero number to the power 0 is 1.

$$\left(\frac{10}{11}\right)^0 = 1$$

**Answer: 1**

# Whiteboards • Question 9 of 9

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Hannah cancels:  $\frac{2^3}{2^3} = \frac{8}{8} = 1$ . Sebastian subtracts:  $\frac{2^3}{2^3} = 2^0$ .

Who is right?

**Write your answer in sentences.**

# Whiteboards • Question 9 of 9

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Hannah cancels:  $\frac{2^3}{2^3} = \frac{8}{8} = 1$ . Sebastian subtracts:  $\frac{2^3}{2^3} = 2^0$ .

Who is right?

Hannah:  $\frac{2^3}{2^3} = \frac{8}{8} = 1$

Sebastian:  $\frac{2^3}{2^3} = 2^{3-3} = 2^0$

Same fraction, so  $2^0 = 1$

**Answer: Both —  $2^0 = 1$**

# Independent Set

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1. Write as one power.  $\frac{6^7}{6^3}$
2. Find the value.  $\frac{5^6}{5^4}$
3. Find the value.  $\frac{(-4)^5}{(-4)^3}$
4. Rewrite so the exponent is on the numerator and on the denominator.  $\left(\frac{3}{7}\right)^4$
5. Find the value.  $\left(\frac{12}{13}\right)^0$
6. Find the value of  $n$ .  $\frac{2^9}{2^n} = 2^4$

# IXL

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1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

## **Today's skills — all required**

Powers with negative bases — WFW

Division with exponents — XNC

Identify equivalent expressions involving whole-number exponents — KBE